SITING EVIDENCE BUNDLE

# Rural water point siting in Mayuge District, Uganda

| | |
|---|---|
| SUBJECT | Placement of 14 additional water points under a fixed budget |
| PREPARED BY | An autonomous analysis agent, from open data, without field verification |
| DECISIONS | All 8 decisions in this run were recorded by a person. |
| REVIEWED BY | Jadey, whose decisions are recorded at Ex07 |
| GENERATED | 2026-09-04T15:59:05Z |
| RUN | `mayuge-water-20260904T155905Z` |
| PROVENANCE | `ac15a4eae9de9c52` |

## Statement

An estimated 570,781 people live in the area of interest. 371,949 of them, 65.2%, live within about 21 minutes of walking over the modelled terrain of a water point recorded as serving. 198,832 people, 34.8% of the district, do not (Ex01, Ex02, Ex03).

Of the 1,251 water point records held for this district, 354 are recorded as serving and 897 are not (Ex01). The median record is 1.9 years old, so the figures above describe the surveyed state of the network rather than its state today (Ex10).

2,095 candidate locations were evaluated (Ex04). Selecting 14 of them by maximum coverage brings a further 76,465 people inside the service radius, raising coverage to 78.6% (Ex05, Ex06). Returns fall away quickly: the first site reaches 12,365 people and the last 3,946, so the number of sites to build is a decision that should be taken on the curve at Ex05 rather than on a round number.

1 planner overrides were applied. They forgo 16,257 people, 3.8% of the coverage the unmodified plan would have reached. Each override, the reason given for it and its individual cost are recorded at Ex07. No override was blocked.

2 of the 10 independent checks at Ex08 returned a flag rather than a pass: data currency, boundary effect. The flagged findings are reproduced in full at Ex08 and are not resolved in this bundle.

3 properties of the source registers would have changed a coverage figure had they been handled differently. Each is recorded at Ex10 with what was observed and what was done about it. None were corrected silently.

*Recommendations in this bundle are advisory. Land availability, tenure, water table, community consent and construction feasibility are outside the data used here and inside the judgement of the officer reviewing it. No figure in this bundle is asserted without an exhibit number after it.*

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## Verification

Every retrieval in this bundle is recorded with its endpoint, the query as executed, the time of retrieval and the record counts before and after cleaning (Ex01, Ex02, Ex11). Cleaning rules are declared per source in the handbook files that accompany this bundle and are applied in order, each reporting its own drop count, so no record leaves the pipeline unaccounted for.

Coverage is recomputed independently of the solver by a separate process before this bundle is produced, and that process can prevent it being produced at all (Ex08). Coverage is always the union over serving locations; a per-location sum double counts every settlement reachable from two of them, and Ex08 reports what that sum would have claimed.

Properties of the source data that would change a figure if read differently are recorded at Ex10 rather than handled silently. Where the agent's own source handbook was found to be incomplete, that is recorded there too.

No figure in this bundle was estimated, interpolated or supplied from memory. A figure is either computed from a recorded retrieval or it is absent.

## Ex00 – Decisions taken, and by whom

**SUPPORTS** That the judgements this assessment rests on are attributable
**CAPTURED** Recorded before the analysis ran
**METHOD** Each row is a judgement about values rather than a technical parameter: what counts as served, whose need the programme answers, whether a register of this age is fit to direct spending. The analysis refuses to run without them rather than assuming one, because assuming one would move the decision away from the officer accountable for it. Where a row is attributed to the agent, no district position was recorded and the agent decided in its absence.

All 8 decisions in this run were recorded by a person.

| Decision | Value | Recorded by | Reason given |
|---|---|---|---|
| `service_radius_m` | 1000.0 metres | Jadey | 1000 m  Mayuge already reports against, |
| `coverage_basis` | walking_time | Jadey | walking time : Mayuge walk around water rather than across it |
| `objective` | max_coverage | Jadey | 198,832 unserved is too many to spend a small programme on the margins |
| `budget` | 10 facilities | Jadey | we have varied budget, we want to find an ideal number |
| `data_currency_accepted` | yes | Jadey | conditional, some will have been repaired and some of the 354 will have failed |
| `coverage_tolerance` | 0.05 share of achievable coverage | Jadey | 198,832 unserved is too many to spend a small programme on the margins |
| `review_floor` | 6.5 weighted score out of 10 | Jadey | don't be too strict |
| `equity_accepted` | yes | Jadey | 198,832 unserved is too many to spend a small programme on the margins |

## Ex01 – Rural water point siting register for Mayuge

**SUPPORTS** The installed stock and its recorded condition
**CAPTURED** https://data.waterpointdata.org/resource/eqje-vguj.json on 2026-09-04, read from the local cache for this run
**METHOD** Socrata query below, executed verbatim. Cleaning rules are declared in handbooks/wpdx.yaml and applied in order; each rule reports its own drop count.

```
clean_country_name = 'Uganda' AND clean_adm2 = 'Mayuge'
```

| | |
|---|---|
| Records returned | 1,378 |
| Records after cleaning | 1,251 |
| Discarded | 127 |

**RECORDS DISCARDED, BY RULE**

| | |
|---|---|
| flagged duplicate by WPdx | 127 |

| | |
|---|---|
| Recorded as serving | 354 |
| Recorded as not serving | 897 |
| Median record age | 1.9 years |

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## Ex02 – Population surface

**SUPPORTS** The denominator for every coverage figure in this bundle
**CAPTURED** https://hub.worldpop.org/rest/data/pop/wpgp on 2026-09-03, read from the local cache for this run
**METHOD** National 100 m grid, read over the area of interest and aggregated into blocks. The aggregation is coarser than the source and finer than the service radius, so it does not bias coverage.

```
UGA ppp 2020, bbox 33.2835 -0.0474 33.7104 0.6061 (WGS 84), aggregated 5x5
```

| | |
|---|---|
| Raster cells in window | 15,912 |
| Aggregated demand cells | 8,992 |
| Population in the area of interest | 570,781 |

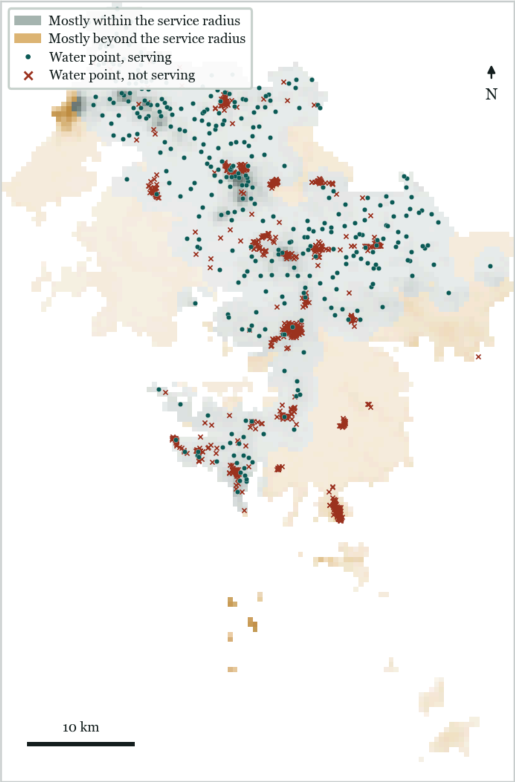
population year 2020; 401,408 source cells at ~100 m read over the window, aggregated 5x5 into 15,912 cells of ~500 m



## Ex03 – Coverage before any intervention

**SUPPORTS** The size of the gap the recommendation is trying to close
**CAPTURED** Computed from Ex01 and Ex02
**METHOD** A demand cell counts as covered when it lies within about 21 minutes of walking over the modelled terrain of a point recorded as serving. Coverage is the union over serving points; a per-point sum would double count every settlement reachable from two of them.

| | |
|---|---|
| Population | 570,781 |
| Covered today | 371,949 (65.2%) |
| Not covered | 198,832 (34.8%) |




*Serving and non-serving points over the population surface. Warm shading is population beyond the service radius.*

## Ex04 – Candidate sites considered

**SUPPORTS** That the recommendation was selected from a stated, reproducible set
**CAPTURED** Constructed from Ex01 and the area of interest
**METHOD** Two kinds of candidate: an existing point recorded as not serving, which could be rehabilitated, and a regular lattice of greenfield locations. Lattice points already inside an existing service area are removed before selection.

| | |
|---|---|
| Candidates evaluated | 2,095 |
| Lattice spacing | 750 m |
| Minimum spacing between recommendations | 1000 m |

## Ex05 – Selection method and its guarantee

**SUPPORTS** That the ranking is reproducible and its optimality is bounded
**CAPTURED** Computed
**METHOD** Candidates are scored by the population they would newly bring inside the service radius and selected greedily. On a monotone submodular coverage function greedy selection is within a factor of 1 - 1/e, about 63 per cent, of the optimum. The exact integer programme is not used in the loop: it takes minutes at district scale where the heuristic takes seconds, and every planner override triggers a re-solve.

| | |
|---|---|
| Sites selected | 14 |
| Population newly covered | 76,465 |
| Coverage after the programme | 78.6% |
| Benchmark against an exact solver | not run (not run) |

| Sites built | Newly covered | Cumulative | Share of district |
|---|---|---|---|
| 1 | 12,365 | 384,314 | 67.3% |
| 2 | 7,240 | 391,554 | 68.6% |
| 3 | 6,715 | 398,268 | 69.8% |
| 4 | 6,100 | 404,368 | 70.8% |
| 5 | 5,076 | 409,444 | 71.7% |
| 6 | 4,775 | 414,219 | 72.6% |
| 7 | 4,609 | 418,828 | 73.4% |
| 8 | 4,589 | 423,417 | 74.2% |
| 9 | 4,533 | 427,950 | 75.0% |
| 10 | 4,207 | 432,157 | 75.7% |
| 11 | 4,167 | 436,324 | 76.4% |
| 12 | 4,093 | 440,417 | 77.2% |
| 13 | 4,051 | 444,468 | 77.9% |
| 14 | 3,946 | 448,414 | 78.6% |

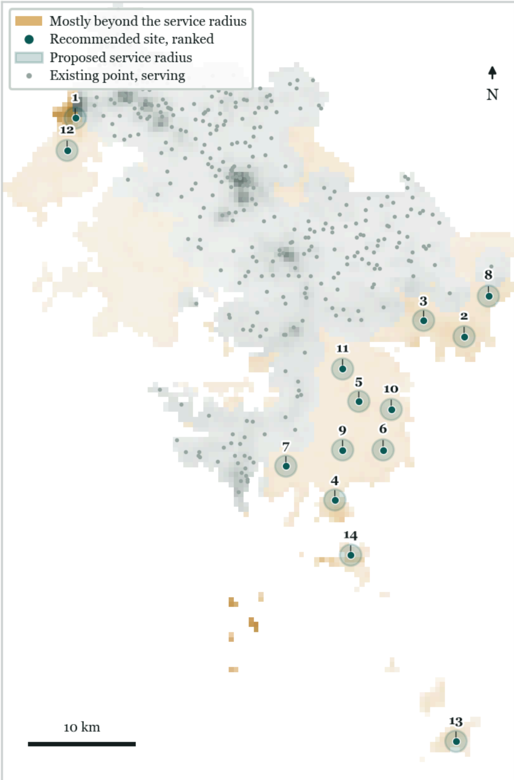
*Marginal and cumulative population covered. The first site reaches 12,365 people, the last 3,946.*

## Ex06 – Recommended sites

**SUPPORTS** The recommendation itself, in coordinates
**CAPTURED** Computed
**METHOD** Sites in selection order. Each row's coverage is additional to every row above it, so the programme must be built from the top down. Coordinates are WGS 84.

| No. | Site | Type | Location (WGS 84) | Newly covered | Nearest serving |
|---|---|---|---|---|---|
| 1 | S-001 | greenfield | 0.5090, 33.3442 | 12,365 | 2.1 km |
| 2 | S-002 | greenfield | 0.3258, 33.6677 | 7,240 | 4.3 km |
| 3 | S-003 | greenfield | 0.3393, 33.6340 | 6,715 | 3.5 km |
| 4 | S-004 | rehabilitate | 0.1889, 33.5601 | 6,100 | 7.5 km |
| 5 | S-005 | greenfield | 0.2715, 33.5800 | 5,076 | 5.4 km |
| 6 | S-006 | greenfield | 0.2308, 33.6003 | 4,775 | 8.4 km |
| 7 | S-007 | greenfield | 0.2172, 33.5194 | 4,609 | 3.0 km |
| 8 | S-008 | greenfield | 0.3597, 33.6879 | 4,589 | 2.8 km |
| 9 | S-009 | greenfield | 0.2308, 33.5666 | 4,533 | 4.8 km |

| No. | Site | Type | Location (WGS 84) | Newly covered | Nearest serving |
|---|---|---|---|---|---|
| 10 | S-010 | greenfield | 0.2647, 33.6070 | 4,207 | 8.5 km |
| 11 | S-011 | greenfield | 0.2986, 33.5666 | 4,167 | 3.8 km |
| 12 | S-012 | greenfield | 0.4819, 33.3374 | 4,093 | 4.5 km |
| 13 | S-013 | greenfield | -0.0135, 33.6609 | 4,051 | 30.7 km |
| 14 | S-014 | greenfield | 0.1426, 33.5733 | 3,946 | 11.8 km |




*Recommended sites and their service radii over the unserved population.*

BASIS FOR EACH SITE

**S-001** New site covering 12,365 people currently beyond the service radius; nearest working point 2.1 km away.

**S-002** New site covering 7,240 people currently beyond the service radius; nearest working point 4.3 km away.

**S-003** New site covering 6,715 people currently beyond the service radius; nearest working point 3.5 km away.

**S-004** Rehabilitate the existing point covering 6,100 people currently beyond the service radius; recorded as non-functional; technology on record: Public Tapstand; nearest working point 7.5 km away.

**S-005** New site covering 5,076 people currently beyond the service radius; nearest working point 5.4 km away.

**S-006** New site covering 4,775 people currently beyond the service radius; nearest working point 8.4 km away.

**S-007** New site covering 4,609 people currently beyond the service radius; nearest working point 3.0 km away.

**S-008** New site covering 4,589 people currently beyond the service radius; nearest working point 2.8 km away.

**S-009** New site covering 4,533 people currently beyond the service radius; nearest working point 4.8 km away.

**S-010** New site covering 4,207 people currently beyond the service radius; nearest working point 8.5 km away.

**S-011** New site covering 4,167 people currently beyond the service radius; nearest working point 3.8 km away.

**S-012** New site covering 4,093 people currently beyond the service radius; nearest working point 4.5 km away.

**S-013** New site covering 4,051 people currently beyond the service radius; nearest working point 30.7 km away.

**S-014** New site covering 3,946 people currently beyond the service radius; nearest working point 11.8 km away.

## Ex07 – Planner overrides and what they cost

SUPPORTS That the recommendation was reviewed by a person, and where it was overruled
CAPTURED overrides YAML, reproduced verbatim below

**METHOD** Each override was applied on its own and the plan re-solved, so every decision carries its own cost rather than one lumped figure. Overrides are never blocked. A veto removes an area, not a single lattice point, because a planner saying not there means a place.

```yaml
# Planner overrides for Mayuge, water domain.
# Every entry needs a reason. The reason is printed in Ex07 alongside what
# the decision cost in coverage.

- verb: RESCOPE
  budget: 14
  actor: Jadey
  reason: >-
    The marginal-coverage curve for the first 10 sites hasn't flattened: site
    10 still adds 4,207 people, close to site 7's 4,609. Before settling the
    capital ask at 10, I want the curve out to 14 to see whether it's worth
    asking for more before this programme closes, per the original budget
    decision's own uncertainty about the ideal number.
```

| Verb | Target | Reason given | Coverage cost |
|---|---|---|---|
| RESCOPE | run scope | The marginal-coverage curve for the first 10 sites hasn't flattened: site 10 still adds 4,207 people, close to site 7's 4,609. Before settling the capital ask at 10, I want the curve out to 14 to see whether it's worth asking for more before this programme closes, per the original budget decision's own uncertainty about the ideal number. | +16,257 |

**Overrides reach 16,257 more people than the unconstrained plan. Greedy selection is not optimal, so a constraint can occasionally improve it.**

## Ex07b – Safety gate decisions

**SUPPORTS** That access, transfer and fallback decisions were taken deliberately
**CAPTURED** Recorded at the point of each decision
**METHOD** A gate never silently permits. Each row is a point at which the run either proceeded and recorded why that was safe, or refused. A refusal removes a source from the analysis and is stated here rather than left for the reader to infer from a missing figure.

| Gate | Outcome | Decided by | Reason recorded |
|---|---|---|---|
| source authorisation | PASS | harness | Water Point Data Exchange Plus (WPdx+) is a public endpoint; no authorisation required |
| source authorisation | PASS | harness | WorldPop gridded population is a public endpoint; no authorisation required |
| guardrail breach | PASS | harness | Overrides reach 16,257 more people than the unconstrained plan. Greedy selection is not optimal, so a constraint can occasionally improve it. |

## Ex08 – Independent checks

**SUPPORTS** That the plan was checked by a process that did not produce it
**CAPTURED** Computed
**METHOD** These checks run in a separate process from the one that produced the plan, can reject it, and do not read the override reasons in Ex07. They check the plan against the data, not against the account given of it. A single rejection stops the bundle being produced.

| Check | Result | Finding |
|---|---|---|
| coverage arithmetic | PASS | union recount matches at 448,414; across the 14 new sites a per-facility sum would report 85,726 against a true union of 84,860, an overcount of 867 |
| geometry | PASS | all 14 sites lie inside the area of interest |
| budget | PASS | 14 sites within the budget of 14 |
| coordinate reference system | PASS | EPSG:32636 (WGS 84 / UTM zone 36N); over 400 sampled pairs the projected distance departs from great-circle by 0.317% at the median and 0.595% at most, or about 6 m on a 1000 m service radius |

| Check | Result | Finding |
|---|---|---|
| data currency | FLAG | median source record is 1.9 years old. The age was put to the accountable officer and accepted: conditional, some will have been repaired and some of the 354 will have failed |
| aggregation sensitivity | PASS | reweighting the demand surface within cells moves the coverage claim from 78.6% to 78.5%, a shift of 0.06% |
| boundary effect | FLAG | 4 of 14 sites lie within one service radius of the district boundary; the nearest is 625 m from it. Population and facilities across the boundary are not represented, so coverage at these sites may be misstated |
| equity | PASS | 53% of newly covered people live in the densest quartile of cells, against 42% district-wide |
| provenance | PASS | 4 sources recorded, all figures traceable |
| cartographic consistency | PASS | 3 figures reviewed against the account; all consistent |

## Ex08b – Scoring review

**SUPPORTS** That the account was scored against a stated floor before issue
**CAPTURED** Computed by a reviewer that reads only this bundle's results file
**METHOD** The reviewer receives the results file and nothing else: not the solver, not the code path, not the override reasons. It cannot score the intention behind the plan, only the account given of it. Every dimension carries a floor and the weighted average carries a floor; the run is issued only when all are met, or when a person forces it and that is recorded.

| Dimension | Score | Floor | Weight | | Basis |
|---|---|---|---|---|---|
| data adequacy | 8.5 | 5.0 | 0.25 | met | median record 1.9 years old, put to the accountable officer and accepted with a recorded reason; 1 semantic anomalies in the register |
| method fitness | 7.5 | 5.0 | 0.20 | met | On a monotone submodular coverage function greedy selection is within a factor of 1 - 1/e, about 63 per cent, of the optimum.; no exact benchmark (not run); reach measured as walking time from the local friction value at each candidate, not as a least-cost traverse |
| spatial rigour | 9.0 | 6.0 | 0.25 | met | 6 of 6 required diagnostics present; distances in EPSG:32636 (WGS 84 / UTM zone 36N) |
| accountability | 10.0 | 6.0 | 0.20 | met | every figure resolves to a recorded retrieval; 3 source anomalies disclosed; 1 overrides recorded with their cost |
| actionability | 9.5 | 4.0 | 0.10 | met | 14 sites with coordinates and a stated basis; marginal coverage curve supports a budget decision; 3 sensitivity scenarios |

| | |
|---|---|
| Weighted score | 8.82 |
| Floor to issue | 6.50 |
| Decision | issue |
| Reviewer | deterministic rules over results.json |

## Ex09 – Sensitivity of the recommendation

**SUPPORTS** How far the recommendation depends on assumptions the planner can change
**CAPTURED** Computed
**METHOD** Each scenario re-solves from the same data with one assumption changed. A recommendation that survives every row is one a planner can defend.

| Scenario | Covered | Share | Against the base case |
|---|---|---|---|
| Rehabilitation of existing points only | 400,081 | 70.1% | -32,076 |
| Half the budget (7 sites) | 418,828 | 73.4% | -13,329 |
| Double the budget (28 sites) | 492,169 | 86.2% | +60,012 |

## Ex10 – Anomalies in the source data, and how they were handled

**SUPPORTS** That the source registers were read rather than assumed
**CAPTURED** Observed during retrieval and cleaning

**METHOD** Every item below is a property of the published data that would change a coverage figure if handled differently. Each is recorded with what was observed, what the agent did, and what a reader must keep in mind. None of these were corrected silently.

**1. SEMANTICS · WPdx+**

| | |
|---|---|
| Observed | status_clean is not two-valued in this district. Values present: Non-Functional (891); Functional, needs repair (354); Functional, not in use (6). |
| Handling | Serving status is read from the status_semantics block in handbooks/wpdx.yaml, which treats Functional and Functional, needs repair as serving and everything else as not serving. |
| Bearing | An equality test against the string Functional would score every point in this district as not serving and inflate the gap. |

**2. DUPLICATION · WPdx+**

| | |
|---|---|
| Observed | 127 of 1,378 returned records were removed: flagged duplicate by WPdx. |
| Handling | Removed before any coverage computation, per the cleaning rules declared in the handbook. |
| Bearing | Coverage computed on the raw response would count these records twice or count superseded observations. |

**3. METHOD · MAP friction surface**

| | |
|---|---|
| Observed | Over this area the walking friction surface implies 21 minutes per kilometre at the median and 60 at the 90th percentile, so the chosen 1000 m corresponds to about 21 minutes of walking. |
| Handling | Coverage is tested as walking time rather than straight-line distance. Reach is computed per candidate from the local friction value rather than by a least-cost accumulation from every candidate, which would be exact but is not tractable at this candidate count. |
| Bearing | Where terrain changes sharply within one facility's reach, the local-friction approximation applies the conditions at the site in every direction, and is conservative rather than optimistic. The surface is modelled at about one kilometre and does not represent local footpaths. |

## Ex11 – Reproduction record

| | |
|---|---|
| **SUPPORTS** | That this bundle can be regenerated from the recorded queries |
| **CAPTURED** | Computed |
| **METHOD** | The bundle is a function of one results file and the manifest that produced it. Any change to a source query, a cleaning rule or a retrieval date changes the provenance hash. |

| | |
|---|---|
| Run | mayuge-water-20260904T155905Z |
| Generated | 2026-09-04T15:59:05Z |
| Provenance hash | ac15a4eae9de9c52 |

```
python -m siting.cli --country Uganda --adm2 Mayuge --iso3 UGA --domain water --mode manual --out sessions/mayuge-20260904T152614Z-6127d5/runs --format both --reviewer rules --decisions sessions/mayuge-20260904T152614Z-6127d5/decisions.yaml --overrides overrides/mayuge.yaml --figure-review sessions/mayuge-20260904T152614Z-6127d5/runs/mayuge-water-20260904T155905Z/figure_review.json --resume mayuge-water-20260904T155905Z
```

| Source | Licence | Accessed |
|---|---|---|
| Water Point Data Exchange (WPdx+), global water point registry, https://www.waterpointdata.org/ | CC BY 4.0 | 2026-09-04 |
| GADM database of Global Administrative Areas, version 4.1, https://gadm.org | GADM licence, free for academic and non-commercial use | 2026-09-03 |
| WorldPop, School of Geography and Environmental Science, University of Southampton, https://www.worldpop.org | CC BY 4.0 | 2026-09-03 |
| Weiss, D.J. et al. Global maps of travel time to healthcare facilities. Nature Medicine 26, 1835-1838 (2020). | CC BY 4.0 | 2026-09-04 |